

HYDROCARBONS WS

Class 11 - Chemistry

- Which of the following reactions of methane is incomplete combustion. [1]

a) $\text{CH}_4 + \text{O}_2 \xrightarrow{\text{Mo}_2\text{O}_3} \text{HCHO} + \text{H}_2\text{O}$

c) $\text{CH}_4 + \text{O}_2 \rightarrow \text{C(s)} + 2\text{H}_2\text{O (l)}$

b) $\text{CH}_4 + \text{O}_2 \rightarrow \text{CO}_2 \text{ (g)} + 2\text{H}_2\text{O (l)}$

d) $2\text{CH}_4 + \text{O}_2 \xrightarrow{\text{Cu/523K/100/atm}} 2\text{CH}_3\text{OH}$
- The boiling point of isomeric branched chain alkene is: [1]

a) lower than the boiling point of straight chain alkene.

c) same boiling point as straight chain alkenes except for optical isomers.

b) higher boiling point than straight chain alkenes.

d) same boiling point as straight chain alkenes.
- Total number of conformation of ethane is: [1]

a) 3

c) 2

b) 4

d) Infinite
- A molecule of benzene has: [1]

a) 6σ and 3π bonds

c) 6σ and 9π bonds

b) 12σ and 3π bonds

d) 9σ and 3π bonds
- In the following sequence of reactions, the alkene is converted to compound *B* [1]

$\text{CH}_3\text{CH} = \text{CHCH}_3 \xrightarrow{\text{O}_3} \text{A} \xrightarrow{\text{H}_2\text{O, Zn}} \text{B}$

The compound *B*----- is?

a) $\text{CH}_3\text{CH}_2\text{CHO}$

c) CH_3COCH_3

b) CH_3CHO

d) $\text{CH}_3\text{CH}_2\text{COCH}_3$
- Alkyl halides react with dialkyl copper reagents to give: [1]

a) Alkenyl halides

c) Alkyl copper halides

b) Alkanes

d) Alkenes
- The treatment of CH_3MgX with $\text{CH}_3\text{C} \equiv \text{C} - \text{H}$ gives _____. [1]

a) $\text{CH}_3 - \text{CH} = \text{CH}_2$

c) $\text{H}_3\text{C}-\text{CH}_3$

b) $\text{CH}_3\text{C} \equiv \text{C} - \text{CH}_3$

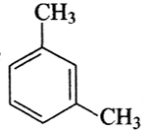
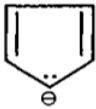
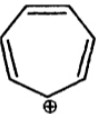
d) CH_4
- Arrange the following hydrogen halides in order of their decreasing reactivity with propene. [1]

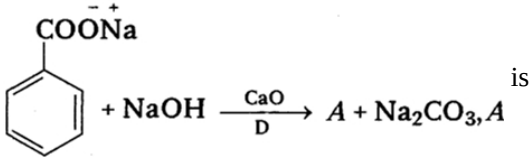
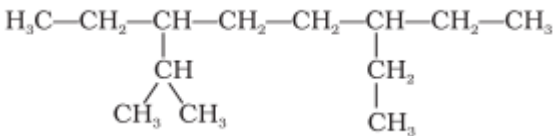
a) $\text{HCl} > \text{HI} > \text{HBr}$

c) $\text{HI} > \text{HBr} > \text{HCl}$

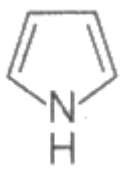
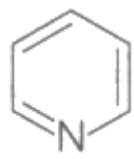
b) $\text{HCl} > \text{HBr} > \text{HI}$

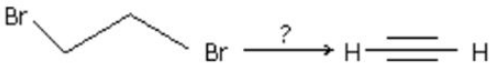
d) $\text{HBr} > \text{HI} > \text{HCl}$

9. Which conformation of ethane has the lowest potential energy? [1]
- a) Eclipsed
b) Skewed
c) Staggered
d) All will have equal PE
10. The following reaction is: [1]
- $$(\text{CH}_3)_3\text{C-Br} \xrightarrow{\text{H}_2\text{O}} (\text{CH}_3)_3\text{C-OH}$$
- a) Displacement reaction
b) Elimination reaction
c) Substitution reaction
d) Free radical reaction
11. Boiling point of alkanes are: [1]
- a) is independent of branching.
b) decreases with increase in molecular mass.
c) is independent of molecular mass.
d) increases with increase in molecular mass.
12. Carbon-carbon double bond in alkenes consists of: [1]
- a) one strong sigma (σ) bond due to head-on overlapping of sp^3 hybridised orbitals one weak pi (π) bond.
b) one strong sigma (σ) bond due to head-on overlapping of sp^2 hybridised orbitals one weak pi (π) bond.
c) one weak sigma (σ) bond due to head-on overlapping of sp^2 hybridised orbitals one weak pi (π) bond.
d) one weak sigma (σ) bond due to head-on overlapping of sp^3 hybridised orbitals one weak pi (π) bond.
13. The IUPAC name of  is [1]
- a) 1,3-dimethylbenzene
b) methyltoluene
c) dimethylbenzene
d) 1,4-dimethylbenzene
14. The aromatic compound among the following is [1]
- a) All of these
b) 
c) 
d) 
15. The number of chain isomers of C_5H_{12} are: [1]
- a) 5
b) 4
c) 2
d) 3
16. Baeyer's reagent is: [1]
- a) K_2MnO_4
b) alkaline KMnO_4
c) bromine water
d) acidified KMnO_4
17. The unusual behavior of benzene is explained by: [1]
- a) Delocalized π electron cloud
b) sp^2 hybridization

- c) sp^3 hybridization
d) Kekule structure
18. Number of σ -bonds and π -bonds present in  is [1]
- a) σ -Bond = 14
 π -Bond = 3
b) σ -Bond = 18
 π -Bond = 4
c) σ -Bond = 15
 π -Bond = 3
d) σ -Bond = 17
 π -Bond = 4
19. An aqueous solution of compound A gives ethane on electrolysis. The compound A is? [1]
- a) Sodium propionate
b) Sodium acetate
c) Sodium ethoxide
d) Ethyl acetate
20.  is [1]
- a) C_6H_6
b) $C_6H_5 - Na$
c) $C_6H_5 - ONa$
d) $C_5H_5 - Na$
21. The compound formed as a result of oxidation of ethyl benzene by $KMnO_4$ is [1]
- a) benzoic acid
b) acetophenone
c) benzophenone
d) benzyl alcohol
22. The correct IUPAC name of the following alkane is: [1]
- 
- a) 3 - Isopropyl - 6 - ethyloctane
b) 5 - Isopropyl - 3 - ethyloctane
c) 3,6 - Diethyl - 2 - methyloctane
d) 3 - Ethyl - 5 - isopropyloctane
23. Which of these is not a characteristic of Aromatic Hydrocarbons? [1]
- a) Carbon atoms form a closed chain or ring.
b) Carbon atoms form a an open chain.
c) They may contain double bonds.
d) They are special type of cyclic compounds.
24. Halogenation of alkanes is a: [1]
- a) Rearrangement reaction
b) Elimination reaction
c) Addition reaction
d) Substitution reaction
25. Which of the following method is not a synthetic method for the production of Benzene? [1]
- a) Decarboxylation of aromatic acids
b) Isolation from coal tar
c) Cyclic polymerization of ethyne
d) By reduction of phenol
26. A dibromo derivative of an alkane reacts with sodium metal to form an alicyclic hydrocarbon. The derivative is _____. [1]
- a) 1, 4-dibromobutane
b) 1, 1-dibromopropane

- c) 2, 2-dibromobutane
d) dibromoethane
27. Vinylcarbinol is: [1]
a) $\text{CH}_3\text{CH}(\text{OH}) = \text{CH}_2$
b) $\text{CH}_3 - \text{C}(\text{CH}_2\text{OH}) = \text{CH}_2$
c) $\text{CH}_3 - \text{CH} = \text{CH} - \text{OH}$
d) $\text{HO} - \text{CH}_2 - \text{CH} = \text{CH}_2$
28. Phenyl magnesium bromide reacts with methanol to give: [1]
a) a mixture of phenol and $\text{Mg}(\text{Me})\text{Br}$.
b) a mixture of anisole and $\text{Mg}(\text{OH})\text{Br}$.
c) a mixture of toluene and $\text{Mg}(\text{OH})\text{Br}$.
d) a mixture of benzene and $\text{Mg}(\text{OMe})\text{Br}$.
29. What does Marsh gas mainly contains? [1]
a) H_2S
b) CH_4
c) C_2H_2
d) CO
30. Which of the following deactivates the benzene ring towards electrophilic substitution and is $\frac{0}{p}$ directing? [1]
a) All of these
b) $-\text{CHO}$
c) $-\text{OCH}_3$
d) $-\text{CI}^\ominus$
31. Presence of a nitro group in a benzene ring: [1]
a) deactivates the ring towards electrophilic substitution.
b) activates the ring towards electrophilic substitution.
c) renders the ring basic.
d) deactivates the ring towards nucleophilic substitution.
32. Toluene reacts with chlorine in the presence of light to give: [1]
a) p-chlorotoluene
b) Benzyl chloride
c) o-chlorotoluene
d) Benzoyl chloride
33. The given structures, $\text{CH}_3 - \overset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ are [1]
a) chain isomers
b) Functional isomers
c) conformational isomer
d) Position isomers
34. Which of the following reactions will yield 2, 2-dibromopropane? [1]
a) $\text{CH} \equiv \text{CH} + 2\text{HBr} \rightarrow$
b) $\text{CH}_3 - \text{CH} = \text{CH}_2 + \text{HBr} \rightarrow$
c) $\text{CH}_3 - \text{C} \equiv \text{CH} + 2\text{HBr} \rightarrow$
d) $\text{CH}_3\text{CH} \equiv \text{CHBr} + \text{HBr} \rightarrow$
35. Which of the following organic materials damage DNA of our body? [1]
a) Tabacco
b) Coal
c) All of these
d) Petroleum
36. The possible rotamers of ethane is/are [1]
a) 2
b) 3
c) ∞
d) 4

37. On passing vapours of phenol over heated zinc dust it gets reduced to [1]
 a) benzene b) toluene
 c) C_6H_5OH d) aniline
38. The IUPAC name of $CH_2 = \underset{\substack{| \\ CH_3}}{C} - CH_3$ is [1]
 a) 2-methylbutene b) 2-methylbut-2-ene
 c) 2-methylprop-1-ene d) 1-methylpropene
39. In the Wurtz reaction method of preparation of alkanes, the stoichiometric coefficient of Sodium is _____. [1]
 a) 5 b) 3
 c) 2 d) 1
40. Benzene reacts with chlorine in the presence of sunlight and in the absence of halogen carriers to give? [1]
 a) $C_6H_6Cl_6$ b) CCl_4
 c) C_6Cl_6 d) C_6H_5Cl
41. The increasing order of reduction of alkyl halides with zinc and dilute HCl is [1]
 a) $R-Cl < R-Br < R-I$ b) $R-I < R-Br < R-Cl$
 c) $R-Br < R-I < R-Cl$ d) $R-Cl < R-I < R-Br$
42. The correct IUPAC name of the given structure $H_3C - \underset{\substack{| \\ CH_3}}{C} H - C \equiv CH$ is [1]
 a) 2-methylbut-1-yne b) 2-methylbut-3-yne
 c) 1-methylbut-3-yne d) 2-methylbut-3-yne
43. Stereoisomeric alkenes due to the different arrangement of atoms or groups in space are referred to as: [1]
 a) Constitutional isomers b) Position isomers
 c) Chain isomers d) Geometric isomers
44. What product is obtained on electrolyzing potassium fumarate? [1]
 a) CH_4 b) C_2H_2
 c) C_2H_4 d) C_2H_6
45. Which of the following compounds is not aromatic? [1]
 a)  b) 
 c)  d) 
46. If degree of unsaturation is three, then a compound shows: [1]
 a) one double bond and two rings b) one ring and two double bonds

- c) one triple bond and one double bond d) all of these
47. Which branched chain isomer of the hydrocarbon with molecular mass 72u gives only one isomer of monosubstituted alkyl halide? [1]
- a) Tertiary butyl chloride b) Neohexane
c) Neopentane d) Isohexane
48. Soda lime is a mixture of [1]
- a) NaOH + Ca(OH)₂ b) NaOH + MgO
c) NaOH + Mg(OH)₂ d) NaOH + CaO
49. Which of these fuels given below is not a hydrocarbon? [1]
- A. Hydrogen
B. LNG
C. LPG
D. CNG
- a) C b) B
c) D d) A
50. Ethene is obtained by electrolyzing: [1]
- a) Pot. Succinate b) Pot. Fumarate
c) Pot. Acetate d) All of these
51. The reagent(s) for the following conversion, is/are [1]
- 
- a) ZN/CH₃OH b) alcoholic KOH
c) aqueous KOH followed by NaNH₂ d) alcoholic KOH followed NaNH₂
52. n-Propylbromide on treatment with ethanolic KOH is: [1]
- a) Propene b) Propanal
c) Propane d) Propyne
53. In cycloalkanes carbon atoms form: [1]
- a) A an open chain b) A zigzag chain
c) A closed chain or a ring d) A open coiled chain
54. The peroxide effect in anti-Markovnikov's addition of HBr to unsymmetrical alkenes involves [1]
- a) homolytic fission of the double bond b) a free radical mechanism.
c) heterolytic fission of the double bond d) an ionic mechanism
55. Which of the following is correct regarding the stability of carbocation? [1]
- a) 3° > 2° > 1° b) 2° > 3° > 1°
c) 2° > 1° > 3° d) 3° < 2° < 1°

56. The organic compound can be formed by: [1]

a) Plants only b) Animals only
 c) Plants, animals, and synthetic methods d) Plants and animals both

57. Which of the following conformations of butane is most stable? [1]

a) Skew b) Eclipsed
 c) Gauche d) Anti

58. When two hydrogen atoms in benzene are replaced by two similar or different monovalent atoms or groups, how many different position isomers are possible? [1]

a) 6 b) 5
 c) 2 d) 3

59. Organic compounds are found in: [1]

a) only vegetable kingdom b) only Animal kingdom
 c) both animal and vegetable kingdom d) only mineral kingdom

60. Alkynes on reduction with sodium in liquid ammonia forms [1]

a) Both cis-alkene and trans-alkene b) cis-alkene
 c) alkane d) trans-alkene

61. Which of the following reactions of methane represents its incomplete combustion? [1]

a) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ b) $\text{CH}_4 + \text{O}_2 \xrightarrow{\text{Mo}_2\text{O}_3} \text{HCHO} + \text{H}_2\text{O}$
 c) $2\text{CH}_4(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}(\text{g}) + 4\text{H}_2(\text{g})$ d) $2\text{CH}_4 + \text{O}_2 \xrightarrow{\text{Cu}, 523\text{K}, 100\text{atm}} 2\text{CH}_3\text{OH}$

62. Acetylene is prepared by the action of water on: [1]

a) all of these b) CaC_2
 c) Silicon carbide d) Al_4C_3

63. The general formula for alkenes is: [1]

a) C_nH_{2n} b) $\text{C}_n\text{H}_{2n+1}$
 c) $\text{C}_n\text{H}_{2n+3}$ d) $\text{C}_n\text{H}_{2n+1}$

64. Addition of cold conc. H_2SO_4 with alkenes is an example of [1]

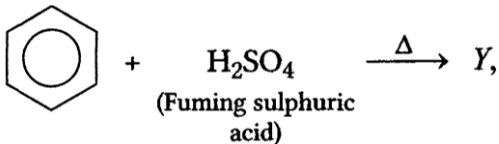
a) nucleophilic addition reaction b) electrophilic substitution reaction
 c) electrophilic addition reaction d) nucleophilic substitution reaction

65. Pyrolysis of coal with heavy oil in the presence of H_2 at 250 atm. and 350°C and catalyst yields. [1]

a) A mixture of low boiling hydrocarbons. b) CH_4
 c) C_3H_8 d) C_2H_6

66. Attack of electrophile on benzene results in the formation of [1]

a) arenium ion b) π Complex

- c) Both σ Complex and arenium ion d) σ Complex
67. The synthesis of 3-octyne is achieved by subsequent stepwise reactions of sodium amide with an alkyne, and a bromoalkane. The bromoalkane and the other alkyne respectively are: [1]
- a) $\text{BrCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{C} \equiv \text{CH}$ b) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{C} \equiv \text{CH}$
- c) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CH}$ d) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CH}$
68. The general formula of a cycloalkane with one benzene ring is _____. [1]
- a) $\text{C}_n\text{H}_{2n-2}$ b) C_nH_n
- c) C_nH_{2n} d) $\text{C}_n\text{H}_{2n+2}$
69. In the reaction, [1]
- 
- a) $\text{C}_6\text{H}_5\text{SO}_3\text{H}$ b) $\text{C}_6\text{H}_5\text{SO}_2$
- c) $\text{C}_6\text{H}_5\text{OH}$ d) $\text{C}_6\text{H}_5\text{CHO}$
70. Among the following, which hydrocarbon is not produced by Wurtz reaction? [1]
- a) Propane b) methane
- c) All of these d) ethane
71. Majority of the reactions of alkynes are examples of: [1]
- a) Electrophilic addition b) Nucleophilic addition
- c) Electrophilic substitution d) Nucleophilic substitution
72. The given reaction, $\text{CH}_3 - \text{Cl} + \text{H}_2 \xrightarrow{\text{Zn, H}^+} \text{CH}_4 + \text{HCl}$ is an example of [1]
- a) dehydrogenation b) carbocation
- c) hydration d) reduction process
73. The carbon atoms of benzene are: [1]
- a) sp^3 - hybridised b) sp - hybridised
- c) sp^2 - hybridised d) non-hybridised
74. Which product is obtained by passing ethanol vapours over heated alumina? [1]
- a) C_2H_2 b) C_2H_6
- c) C_2H_4 d) CH_4
75. Which of the following has higher dipole moment; cis-2-butene and trans-2-butene? [1]
- a) trans-but-2-ene b) Both have different dipole moment
- c) cis-but-2-ene d) Both have same dipole moment

